Ravi Mukkapati

AI Data Engineer | Cloud Data Architect | GCP, OCI, AWS

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PROFESSIONAL SUMMARY

AI Data Engineer and Cloud Data Architect with 20+ years of experience designing data platforms and production pipelines on GCP, OCI and AWS for healthcare, pharma and financial services. Builds the data foundation that AI systems depend on: Python and PySpark ETL/ELT, HL7 ingestion, Snowflake and BigQuery warehousing, and governed, validated data delivery. Hands-on with LLM agents (Google ADK), Model Context Protocol (MCP) servers using FastMCP and Snowflake Cortex, retrieval-augmented generation (RAG) with vector databases, and ONNX model deployment. Postgraduate in AI and Machine Learning from the University of Texas at Austin. Experienced in regulated environments, with PII masking, encryption and data quality controls.

TECHNICAL SKILLS

AI & LLM Engineering: LLM Agents, Google ADK (Agent Development Kit), Model Context Protocol (MCP), FastMCP, Snowflake Cortex Analyst, Snowflake Agents, Retrieval-Augmented Generation (RAG), Vector Databases, Multi-Agent Frameworks (foundational), ONNX, FastAPI, Streamlit

Machine Learning: Scikit-learn, XGBoost, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Vertex AI, SageMaker, OCI Data Science

Programming: Python, PySpark, SQL, PL/SQL, Shell, Scala, Java, HiveQL, Pandas, NumPy, boto3

Data Engineering: ETL/ELT, Change Data Capture (CDC), Batch and API Data Pipelines, HL7 v2, Parquet, Data Modeling, dbt, Fivetran, Informatica, Oracle GoldenGate, ODI, AWS DMS, Apache Sqoop, Erwin, SqlDBM

Orchestration: Apache Airflow, Prefect, Control-M, AWS Step Functions, dbt

Cloud - GCP: BigQuery, Dataflow, Dataproc, Vertex AI, Cloud Storage, Cloud Functions, Compute Engine

Cloud - OCI: Object Storage, Data Integration, Data Flow, Functions, Data Science, Autonomous Data Warehouse (ADW), Autonomous Transaction Processing (ATP)

Cloud - AWS: S3, Glue, Athena, EMR, Lambda, ECS, EC2, DMS, SageMaker, Step Functions, SQS, SNS, IAM

Data Platforms: Snowflake, BigQuery, Oracle, PostgreSQL, MySQL, SQL Server, MongoDB, DuckDB, Hadoop Hive

DevOps & IaC: Terraform, CloudFormation, Docker, GitHub, GitLab, Jenkins, Ansible, Azure DevOps, Liquibase, CI/CD, Scrum

Governance & Security: Data Quality, Data Reconciliation, Data Masking, Encryption, PII Protection, Data Governance

BI & Visualization: Tableau, Power BI, QuickSight, Streamlit

Legacy Platforms: Oracle PeopleSoft (HCM, FSCM, CRM), Oracle BI Apps, OBIEE

PROFESSIONAL EXPERIENCE

Baptist Health South Florida

Cloud Data Architect **08/2019 - Present**

- Built a production support agent with Google ADK on GCP that monitors Airflow DAGs during a cloud migration, triages failures, surfaces SLA risks, and generates operator-ready remediation summaries.
- Integrated MCP servers with Snowflake Cortex Analyst and Snowflake Agents to enable governed natural-language analytics, automate metadata discovery, and speed up recurring data investigations.
- Developed MCP tool servers with FastMCP and FastAPI to expose data services to LLM agents.
- Implemented retrieval-augmented generation (RAG) workflows on vector databases to enable semantic search and grounded answers over enterprise data.
- Deployed trained machine learning models in ONNX format for portable production inference.
- Engineered 20+ production ETL/ELT pipelines in Python and PySpark on AWS (S3, ECS, Lambda), loading curated datasets into Snowflake and cutting data refresh latency by ~35%.
- Designed and built the FL HIE HL7 v2 ingestion pipeline that parses MSH, PID, PV1, OBR and OBX segments and normalizes patient, encounter and observation data into Parquet for scalable Snowflake loading.
- Automated validation and reconciliation controls across the HL7-to-Snowflake workflow (required-field checks, source-to-target mapping validation, duplicate detection, record-count balancing), improving load reliability and reducing downstream data defects.

- Orchestrated pipelines with Apache Airflow, dbt, Control-M, AWS Glue and DMS for full and incremental loads, MERGE logic, soft deletes and historical archival; designed DAGs that move files from source systems to the S3 data lake and Snowflake.
- Built a Python and Apache Spark API ingestion framework with FastAPI and Postman, integrating Zoom, ServiceNow, Salesforce, Credential Stream and Midas data into the platform.
- Delivered end-to-end Provider Data Management solutions that significantly reduced data errors and improved data quality.
- Provisioned cloud infrastructure (DMS, ECS, EMR, Snowflake) with Terraform and implemented change data capture with Oracle GoldenGate and AWS DMS.

ERP Application Architect **02/2016 - 08/2019**

- Architected and delivered Oracle PeopleSoft ERP modules (HCM, ELM, CRM, FSCM) and Oracle BI Apps for a multi-hospital healthcare system, including data migration from Lawson.
- Led enterprise integrations (payroll, benefits, ELM, vendors) and data migrations during hospital acquisitions using files, XML and RESTful APIs.
- Designed BI reports and dashboards on AWS S3, Glue and Athena, and implemented CI/CD workflows with Apache Airflow DAGs.
- Implemented data safeguarding for PII, including Oracle data masking, encryption and sensitive-data hiding.
- Tuned SQL performance, implemented data archival, and mentored engineers through design and code reviews.

EARLIER EXPERIENCE

Application Development Lead, JPMorgan Chase (Chase Paymentech)	2013 - 2015
Oracle Consultant, Int Info Tech	2009 - 2012
Technical Lead, 8x8 Inc	2008 - 2009
Project Lead, IBM, Pune, India	2006 - 2007
Software Engineer, TCS, Bangalore, India	2004 - 2005

SELECTED PROJECTS

AI and Data Platform Projects | Baptist Health South Florida, Cloud Data Architect

AI Support Agent for Airflow on GCP | *Google ADK agent*
Challenge. Production Airflow DAGs were moving to GCP, and failures and SLA risks needed continuous monitoring during the migration. **Solution.** Built an agent with Google ADK that watches DAG runs, triages failures, flags SLA risks, and writes an operator-ready remediation summary for each issue. *Tech: Google ADK, Airflow, GCP, Python.*

Governed Natural-Language Analytics | *Snowflake Cortex and MCP*
Challenge. Recurring data investigations and metadata discovery depended on manual work. **Solution.** Integrated MCP servers with Snowflake Cortex Analyst and Snowflake Agents so users can ask questions in natural language while access stays governed, and automated metadata discovery. *Tech: Snowflake Cortex Analyst, Snowflake Agents, MCP, FastMCP.*

FL HIE HL7 v2 Ingestion Pipeline | *Healthcare interoperability*
Challenge. Segment-based HL7 v2 messages from the FL HIE had to become analytics-ready data in Snowflake. **Solution.** Parsed MSH, PID, PV1, OBR and OBX segments, normalized patient, encounter and observation data, and wrote optimized Parquet files for scalable Snowflake loading. **Result.** Automated validation and reconciliation (required-field checks, source-to-target mapping validation, duplicate detection, record-count balancing) improved load reliability and reduced downstream defects. *Tech: Python, HL7 v2, Parquet, Snowflake.*

AWS-to-Snowflake Data Platform | *Pipelines at scale*
Challenge. Data from many source systems and flat files had to land reliably in Snowflake through both full and incremental loads. **Solution.** Engineered 50+ ETL/ELT pipelines in Python and PySpark on S3, ECS, Lambda, Glue and EMR, orchestrated with Airflow, dbt and Control-M, with Terraform-provisioned infrastructure and change data capture through GoldenGate and DMS. **Result.** Cut data refresh latency by about 35%. *Tech: Python, PySpark, AWS, Airflow, dbt, Terraform, Snowflake.*

Multi-System API Ingestion Framework | *Enterprise integrations*
Challenge. Zoom, ServiceNow, Salesforce, Credential Stream and Midas each exposed their data through separate APIs. **Solution.** Created a common Python and Apache Spark framework, and used FastAPI and Postman to develop, test and productionize the API pipelines. *Tech: Python, Spark, FastAPI, Postman.*

Earlier Enterprise Projects

ERP Modernization: Lawson to PeopleSoft | *Baptist Health South Florida, ERP Application Architect*
Challenge. A multi-hospital healthcare system was moving from Lawson to PeopleSoft ERP (HCM, ELM, CRM, FSCM, Portal) alongside Oracle BI Apps. **Solution.** Gathered requirements from HRIS, Payroll, Benefits and BI teams, then led payroll, benefits, ELM and vendor integrations (files, XML, REST) and data migrations as hospitals were acquired. Built employee and manager self-service pages, custom BI letters, and Learning Management features including external-user enrollment and payment for clinical courses offered through AHA. Implemented Oracle data masking, encryption and sensitive-data hiding to reduce PII exposure.
Tech: PeopleSoft, Oracle, Integration Broker, OBIEE, AWS Glue and Athena.

Payment Processing Applications | *JPMorgan Chase (Chase Paymentech), Application Development Lead*
Challenge. Support credit, debit, gift, prepaid and ACH payments online and at point of sale, alongside an ERP holding critical merchant and customer data. **Solution.** Led cross-functional teams and third-party vendors, built XML and SOA web-service integrations, designed batch processing, data archival and field-level data security, wrote shell scripts for encrypted file transmission, and tuned Oracle database objects for performance.
Tech: PeopleSoft CRM, Oracle, Control-M, Git, Shell, XML.

Pharma Data Warehouse and Commercial BI | *Affymax/Takeda Pharmaceuticals, BI Consultant (Int Info Tech)*
Challenge. Bring Takeda, IMS drug-distribution, Salesforce and call data into one data warehouse for executive and business users. **Solution.** Led requirements sessions with executive management, built Informatica ETL from those sources, and designed OBIEE 11g dimension hierarchies with data-level security. Added Navteq maps for ZIP-level sales and integrated Salesforce Veeva to show real-time account, contact and call data.
Tech: Informatica, OBIEE 11g, Oracle, DAC, Salesforce Veeva.

Equipment and Procurement Data Warehouse | *Asplundh Tree Expert, Data Analytics Consultant (Int Info Tech)*
Challenge. Extend an Equipment Management data warehouse to fit the client's business model. **Solution.** Built Crystal Reports for asset depreciation, disposal and transfer, extended the Vouchers interface for FedEx and telecom data, corrected invoice values for vendors with missing contacts, and delivered OBIEE procurement and spend dashboards.
Tech: PeopleSoft FSCM, OBIEE, Crystal Reports, Oracle.

CRM Integrations and Migrations | *8x8, Panasonic and Apple*

- 8x8 (VoIP provider), Technical Lead: built synchronous APIs for real-time sync with DSS, FAP and Finance, and integrations with the Level3 provider and shippers.
- Panasonic Electronics (IBM), Project Lead: built synchronous and asynchronous integrations with Integration Broker and XML/XSLT, and designed the product registration process.
- Apple (TCS), Software Engineer: helped migrate Apple's Vantive CRM to PeopleSoft and built localized self-service and agreement pages for APAC and European regions.

CERTIFICATIONS

- AWS Certified Solutions Architect
- AWS Certified Cloud Practitioner
- SnowPro Certified, Snowflake
- Oracle Certified Expert

EDUCATION

Postgraduate Program in Artificial Intelligence and Machine Learning, University of Texas at Austin 2023 - 2024
Master's Degree in Computers, Bharathidasan University 2000 - 2003